higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY
(Second Paper)
NQF LEVEL 2

7 NOVEMBER 2019

SYMBOLS	EXPLANATIONS
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG/RD/RM	Reading from table/graph/drawing/document/map
F	Choosing correct formula
SF	Substitution in formula
MF	Manipulation of formula
R/J	Reasoning/Justification
P	Penalty, for example, no units, incorrect rounding, et cetera
R	Rounding off
E	Explanation

This marking guideline consists of 8 pages

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QUESTION 1 (NOTE: Do not subtract marks for "R" omitted)

1.1	1.1.1	**Income** / **Expenses** Stipend R1 800 — **Variable expenses:** Allowance R800 — Groceries ✓R 1100 Entertainment ✓R250 Haircut ✓R60 Airtime and data ✓R120 **Total variable expenses R1 530** **Fixed expenses:** Taxi fare ✓R300 Clothing account ✓R150 Cellphone contract (flat rate) ✓R110 Rent ✓R550 **Total fixed expenses R1 110** **Total income ✓R2 600** — **Total expenses ✓R2 640**	4 Var Exp (deduct 1 for incorrect item) 1 Total Income 4 Fix Exp (deduct 1 for incorrect item) 1 Tot Exp	(10)
	1.1.2	Variable expenses can vary in value from one month to the next. ✓ It is important to know what expenses may vary and also by how much they may vary so that you can prepare in case they increase. ✓ (Accept any other suitable answer)	1 E 1 R/J	(2)
	1.1.3	Themba's budget has a deficit of R40 (R2 600 – R2 640) which is not a good thing since his expenses are more than his income. ✓ This means that he will not be able to afford some expenses. ✓(Accept any other suitable answer)	2 E	(2)
	1.1.4	Themba should reduce the amount spent on any of the variable✓ expenditure items since they are not essentials, such as money spent on entertainment, airtime and data✓etc. (Accept any other suitable answer)	1 J/R 1 E	(2)
1.2	1.2.1	March. ✓	1 A	(1)
	1.2.2	R311,30 + R170,00 + R116,54 ✓ = R597,84✓	1 M 1 A (2 Answer only)	(2)

	1.2.3	R257,20✓	1 A	(1)

	1.2.4	Yes. ✓ R800,00 was paid on 28 March (28/03/2019). ✓	1 A 1 E	(2)

	1.2.5	"Amount Due" = "Amount due from last month" + "Card Total" – "Payment received" = R257,20 + R1 944,94 – R800,00✓ = R1 402,14✓ (No marks for answer only)	 1 M 1 A	(2)

1.3	Price excluding VAT = $\frac{Price\ \ icluding\ \ VAT}{1,15}$ = $\frac{R\,24,69}{1,15}$ ✓ = R21,47 ✓ VAT = original price – price without VAT = R24,69 – R21,47✓ = R3,22 ✓	1 M 1 A 1 M 1 CA	(4)

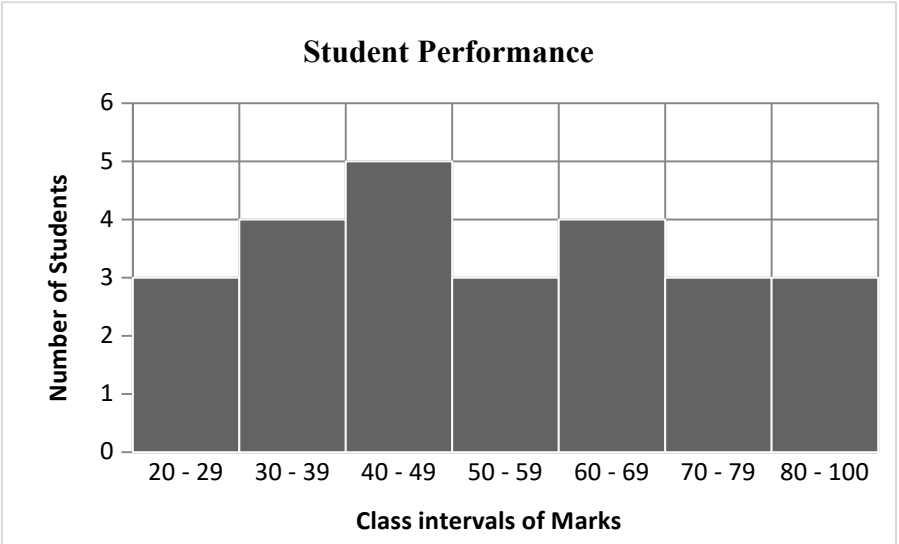
1.4	1.4.1(a)	Total deductions = Tax + Med Aid + Pension + UIF ✓ = R2 112,50 + R2 849,20 + R2 177,50 + R346,60 = R7 485,80 ✓ (Note: No marks for answer only)	1 M 1 A	(2)

	1.4.1(b)	Net Pay = Gross pay – Total deductions ✓ = R34 660,00 – R7 845,80 = R27 174,20 ✓(Note: No marks for answer only)	1 M 1 A	(2)

	1.4.2	Increase = 8% × R32 500,00 = R2 600,00✓ New basic salary = R2 600,00 + R32 500,00 ✓ = R35 100,00 ✓	 1 A 1 M 1 A (2 Answer only)	(3)

	1.4.3	New Gross = New Basic + Overtime = R35 100 + R2 160,00 = R37 260,00✓ New UIF = 1% × R37 260,00 ✓ = R372,60✓ New Net Pay = R37 260,00 – (R2 112,50 + R234,00) – R2 849,20 – R2 177,50 – R372,60 ✓ = R37 260,00 – R7 745,80 = R29 514,20 ✓ (No mark for R29 514,2)	 CA 1.4.2 1 A 1 M 1 A 1 M (CA UIF) 1 CA	(5)
				[40]

QUESTION 2

2.1	2.1.1	$Mean = \frac{(27+26+25+31+29+28+33+28+26)}{9}$ ✓ $= \frac{226}{9}$ ✓ $= 28{,}25$ ✓	1M 1A 1S (3 Answer only)		(3)
	2.1.2	$\{25;\ 26;\ 26;\ 27;\ 28;\ 28;\ 29;\ 31;\ 33\}$ $Median = 28$	2 Arranging data 1 determining median (3 Answer only)		(3)
	2.1.3	$Mode = 28$ ✓✓	2RT		(2)
	2.1.4	$Range = 33 - 25 = 8$	2MA (2 Answer Only)		(2)
2.2	2.2.1	 2 marks for intervals on horizontal axis ✓✓ 1 mark for the following correct bars: 20 – 29 and 30 – 39 ✓ 40 – 49 and 50 – 59 ✓ 60 – 69, 70 – 79 and 80 - 89 ✓ Vertical axis: number of students ✓ Horizontal axis: class intervals of marks ✓ 2 marks for histogram ✓✓ (Note: If a bar graph is drawn: max 4 marks)	3 for correct bars 1 labeling Vertical axis 1 labeling Horisontal axis 2 marks for intervals on horizontal axis 2 marks for histogram		(9)

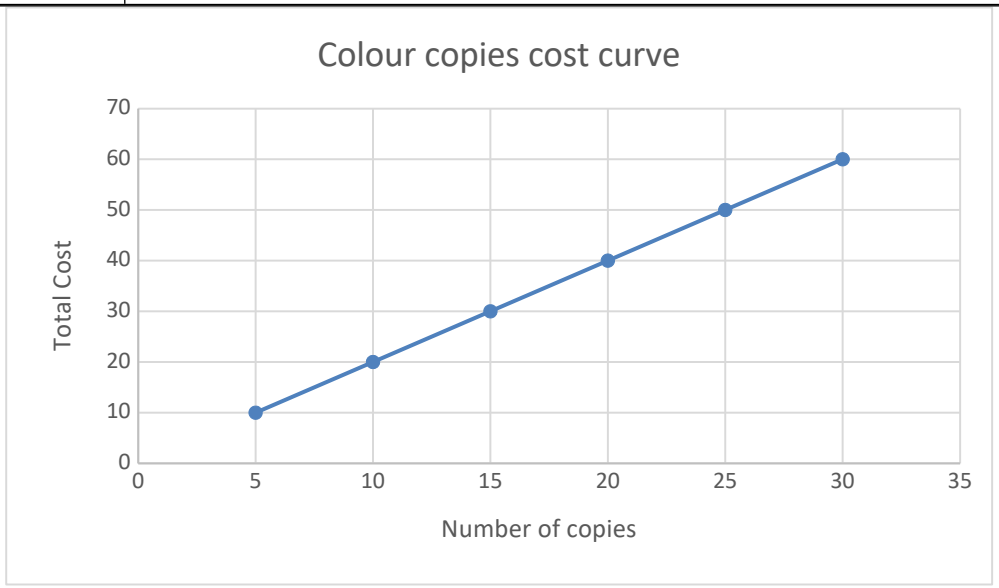
	2.2.2	The 40 – 49 class interval ✓✓	2RT	(2)

2.3	2.3.1	(210+ 260 + 290 + 350 + 360 + 390 + 390 + 380 + 350 + 340 + 320 + 300) ✓÷ 12✓ = R3 940✓ ÷ 12 = R328,33✓ Do not subtract marks for "R" omitted	1 RG 1 M 1 A 1 CA (4 Answer only)	(4)
	2.3.2	Knowing the average monthly cost of electricity helps this household not to budget too little for the monthly electricity cost, ✓✓ or to over-spend on other household finances. (Accept any other suitable answer)	2 E	(2)
	2.3.3	The cost of electricity in January, ✓ February, ✓ March, ✓ November and December✓ is below the average monthly cost, the costs in April, ✓ September and October✓ are slightly higher or more than the average monthly cost. ✓	7 E	(7)
	2.3.4	The most likely reason for the above trend is that electricity usage increases in the winter months in South Africa. ✓(Accept any other suitable answer)	1 E	(1)
				[35]

QUESTION 3 (Note: Do not subtract marks for "R" omitted)

3.1	3.1.1	A = 10✓✓ B = 15✓✓ C = 60✓✓	2MA 2MA 2MA (6 Answers only)		(6)
	3.1.2	Directly proportional.✓As the number of copies increase, so does the cost.✓	2R/J		(2)
	3.1.3 (a)	Dependent variable = Total cost ✓ ✓ (R)	2 A		(2)
	(b)	Independent variable= No. of copies ✓✓	2 A		(2)

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	3.1.4	 Title: ✓ Vertical axis: ✓Horizontal axis: ✓ Straight line drawn ✓ For each 3 correct coordinates: ✓✓	1 Title 2 Axis 3 Graph		(6)
	3.1.5	$No\ \ copies = \frac{122}{2}$✓ $= 61$✓	1M 1A (2 Answer only)		(2)
3.2	3.2.1	Fuel = 50 – 45 or 45 – 40✓ etc. = 5 ✓ℓ	1 M 1 A (2 Answer only)		(2)
	3.2.2	Indirectly/inversely proportional.✓As the distance travelled increases, ✓ the amount of fuel remaining decreases.✓	3R/J		(3)
	3.2.3	A = 40✓ B = 250 ✓ C = 20✓ D = 450✓	4RG		(4)
	3.2.4	Number of litres = 1 000 km✓ ÷ 10 km/litre✓ = 100 ✓litres	2 SF 1 A (3 Answer only)		(3)
	3.2.5	Number of litres = total amount ÷ price per litre = R920,15 ✓÷ R16,73✓	1M 1S 1A		

		= 55✓ litres	(3 Answer only)		(3)
					[35]

QUESTION 4 (Note: Do not subtract marks for "R" omitted)

4.1	4.1.1	*Area* = 16 × 12 = 192 m^2	2 S 1 A (3 Answer only)		(3)
	4.1.2	*Perimeter* = (2 × 16) + (2 × 12) = 56 *m*	1 S 1 A with unit (2 Answer with unit only)		(2)
	4.1.3	*Apartment* = 4100 ÷ 192 = 21,35 = 21	CA 4.1.1 2 MA 1 R (3 Answer only)		(3)
	4.1.4	*Cost* = *R*268 000 × 21 = *R*5 628 000	CA 4.1.3 2 MA (2 Answer only)		(2)
	4.1.5	*Profit* = (450 000 – 268 000) × 21 = *R*182 000 × 21 = *R*3 822 000	2 M 2 CA		(4)
4.2	95 cm = 0,95 m Length of each diagonal is as follows: (length)^2 = (0,95 m)^2 + (1,2 m)^2 = 0, 9025 m^2 + 1,44 m^2 = 2,34 m^2 Therefore, length = $\sqrt{2{,}34\,m^2}$ = 1,53 m Total length of steel = 2 × 1,2 m + 2 × 0,95 m + 2 × 1,53 m = 7,36 m Since the steel is usually sold per metre, he should buy 8 m of steel. <u>or</u> (length)^2 = (95 cm)^2 + (120 cm)^2 = 9 025 cm^2 + 14 400 cm^2 = 23 425 cm^2 Therefore, length = $\sqrt{23\ 425\ cm^2}$ = 153,05 cm Total length of steel = 2 × 95 cm + 2 × 120 cm + 2 × 153,05 cm = 736,1 cm = 7,36 m	1 C 1 M 1 A with rounding 1 A 1 M 1A 1 E 1 M 1 A 1 A 1 M 1 A 1 C with rounding		(7)

	Since the steel is usually sold per metre, he should buy 8 m of steel. or = 8 m	1 E			

4.3	4.3.1	Volume = length × breadth × height = 5 m × 5 m × 50 m✓ = 1 250✓ m^3 ✓	1 SF 1 A 1 U (3 Answer only)		(3)
	4.3.2	$V_B = 3{,}14 \times (0{,}25)^2 \times 1$ $= 0{,}196\ m^3$	2SF 1A (3 Answer only)		(3)
	4.3.3	Volume of rectangular tank = 1 250 × 1 000 *l* = 1 250 000 *litres* Volume of cylinder bucket = 0,196 × 1 000 *litres* = 196 *litres* 75% volume of tank = 0,75 × 1 250 000 *litres* = 937 500 *litres* Number of cylinder bucket = 937 500 ÷ 196 = 4 783,163 = 4 783 full buckets	1 M (CA 4.3.1) 1 C (CA 4.3.2) 1 C 1 A 1 M 2 CA		(7)
4.4	4.4.1	Lusaka and Abuja	2RG		(2)
	4.4.2	*Distance on map* = 29 mm (accept 28 – 30 mm) *Actual distance* = 29 × 69 = 2 001 *km*	1 Measurement 2 M 1 A (4 Answer only)		(4)
					[40]

		TOTAL:		**150**